

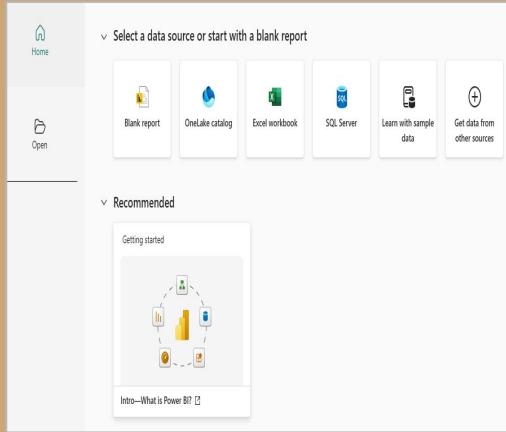
Detecting Anomalies Detecting Anomalies in Time-Series Data Using Power BI

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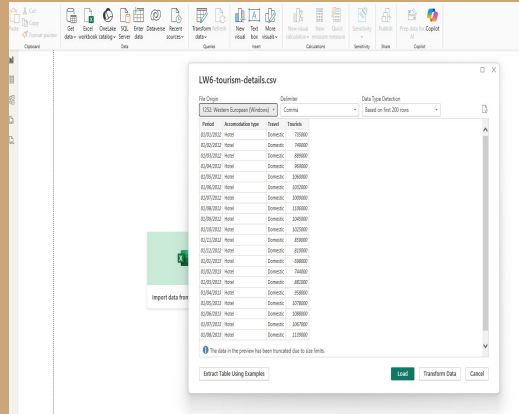
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IS 107 - Business Intelligence

Part 1: Data Preparation



1. Download the dataset and open Power BI Desktop.



2. Click Get Data → Text/CSV and import the dataset.

The screenshot shows the 'LWS-tourism-details' table in Power BI Desktop. The table has four columns: Period, Accommodation type, Travel, and Tourists. The data is sorted by Period in descending order.

	Period	Accommodation type	Travel	Tourists
1	01/01/2021	Hotel	Domestic	720000
2	01/02/2021	Hotel	Domestic	740000
3	01/03/2021	Hotel	Domestic	880000
4	01/04/2021	Hotel	Domestic	960000
5	01/05/2021	Hotel	Domestic	1060000
6	01/06/2021	Hotel	Domestic	1020000
7	01/07/2021	Hotel	Domestic	1000000
8	01/08/2021	Hotel	Domestic	1100000
9	01/09/2021	Hotel	Domestic	1040000
10	01/10/2021	Hotel	Domestic	1020000
11	01/11/2021	Hotel	Domestic	850000
12	01/12/2021	Hotel	Domestic	830000
13	01/01/2022	Hotel	Domestic	690000
14	01/02/2022	Hotel	Domestic	740000
15	01/03/2022	Hotel	Domestic	882000
16	01/04/2022	Hotel	Domestic	950000
17	01/05/2022	Hotel	Domestic	1019000
18	01/06/2022	Hotel	Domestic	1000000
19	01/07/2022	Hotel	Domestic	1067000
20	01/08/2022	Hotel	Domestic	1130000
21	01/09/2022	Hotel	Domestic	1050000
22	01/10/2022	Hotel	Domestic	1044000
23	01/11/2022	Hotel	Domestic	897000
24	01/12/2022	Hotel	Domestic	852000
25	01/01/2023	Hotel	Domestic	740000

3. Ensure the following fields are correctly assigned

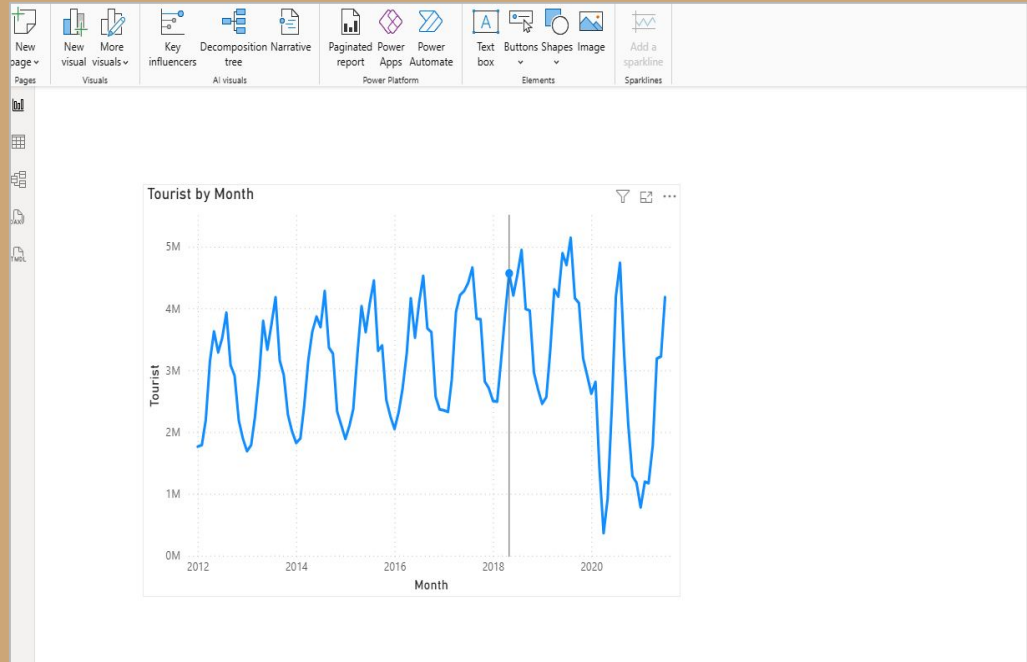
Part 2: Data Visualization

4. Insert a Line Chart visualization.

5. Drag:

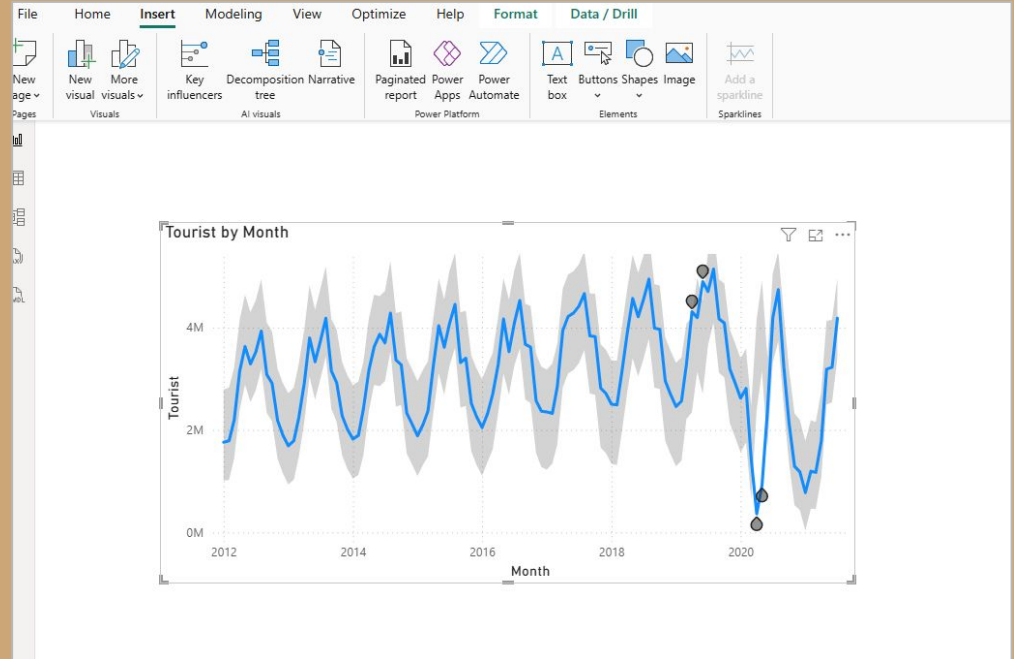
- Period → Axis
- Tourists → Values

6. Expand the date hierarchy to show monthly data.



Part 3: Enable Anomaly Detection

7. Go to the Analytics Pane.
8. Click Find Anomalies → Add.
9. Observe:
 - Highlighted anomaly points
 - Expected value range (shaded region)



Part 4: Explain Anomalies

10. Drag the following into Explain By:

- Accommodation Type

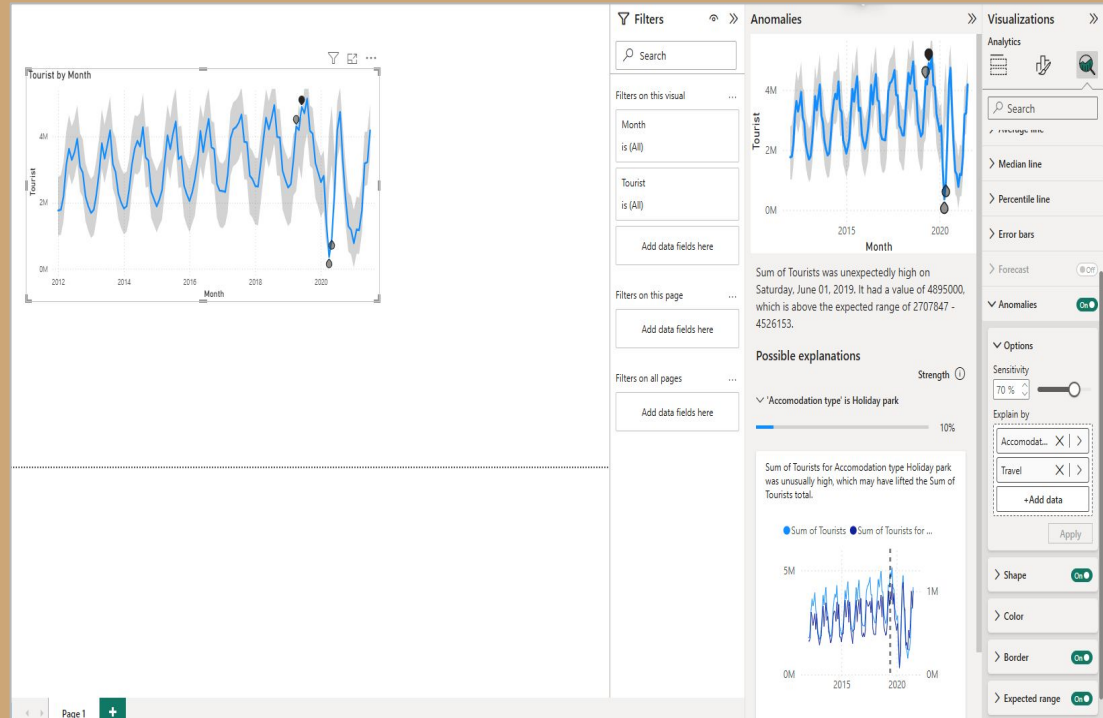
- Travel

11. Click on any detected anomaly.

12. Analyze the Anomalies Pane:

- Identify whether the anomaly is high or low

- Review suggested explanations



Part 5: Interpretation and Analysis

1. What trend and seasonality patterns are observed?

- **Trend:** Gradual rise from 2012–2019, sharp drop in 2020, followed by partial recovery.
- **Seasonality:** Clear annual cycles with regular peaks and troughs, showing predictable seasonal fluctuations.

2. Which anomalies are obvious vs. subtle?

- **Obvious:** Extreme low in 2020–2021 and extreme high in June 2019, both far outside expected ranges.
- **Subtle:** Smaller deviations in earlier years, noticeable against expected bands but less dramatic.

3. What factors contributed to anomalies?

- 2019 high: Unusually high number of tourists choosing holiday park accommodation.
- 2020–2021 low: Likely linked to global travel disruptions (e.g., COVID-19 pandemic).

4. Are the explanations logically valid? Why or why not?

- The 2019 explanation is logically sound but limited—holiday parks contributed only 10% of the anomaly, so other factors remain unaccounted for.
- The 2020–2021 pattern aligns with well-known real-world events, making it highly valid.

Part 6: Enhancement Task

14. Add your own new anomaly dataset (sample below) into your existing dataset.

15. Refresh the report and observe changes.

16. **Compare:**

- Before vs. after anomaly detection
- Newly detected anomalies

